TCP/IP Suite Theory

1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Identify Network Layers Using Wireshark:-

Steps to Perform:-

1. Open **Wireshark** on your laptop.
2. Select your active **Wi-Fi or Ethernet** interface.
3. Click **Start Capture**.
4. Open your browser and refresh the **Flipkart homepage**.
5. Return to Wireshark and click **Stop**.
6. Select packets and expand their headers in the **Packet Details** panel.

Layer	Protocol	Information
Application	HTTPS/TLS	Web communication
Transport	TCP	Source and destination ports
Internet	IP	Source and destination IP addresses
Network Access	Ethernet	Source and destination MAC addresses

2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Real-World Example
Application	Application	Sending a WhatsApp message
Presentation	Application	Encrypting a WhatsApp message
Session	Application	Maintaining a WhatsApp call session
Transport	Transport	TCP/UDP transfers message data
Network	Internet	IP address routes data to the receiver
Data Link	Network Access	Wi-Fi transfers data to the router
Physical	Network Access	Wireless signals carry data between devices

3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

TCP Header Field	Function
Source Port	Identifies the sender's application.
Destination Port	Identifies the receiver's application.
Sequence Number	Shows the position of data in the segment.
Acknowledgment Number	Indicates the next byte expected from the receiver.
Header Length	Specifies the size of the TCP header.
Flags	Controls communication, such as SYN, ACK, and FIN.
Window Size	Indicates how much data the receiver can accept.
Checksum	Checks the TCP segment for transmission errors.
Urgent Pointer	Identifies urgent data when the URG flag is set.
Options	Carries additional TCP settings and information.

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

Explanation of Each Layer

1. **Application Layer:** The user types a message in the Instagram app.
2. **Transport Layer:** TCP helps ensure reliable delivery, manages the data segments, and uses port numbers.
3. **Internet Layer:** IP addresses are added so routers can forward the data toward the destination.
4. **Network Access Layer:** Data is placed into frames and transmitted through Wi-Fi or mobile data.

5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Wireshark Packet Analysis:-

1. Selected Packet: TCP Packet

2. TCP/IP Layer: Transport Layer

3. How I Identified It:
I identified the packet by checking the packet details in Wireshark. The packet shows TCP header information, such as Source Port, Destination Port, and Sequence Number.

4. Real-World Action:
When I refresh the Flipkart homepage, my device communicates with the server to load the website. TCP helps establish a reliable connection and ensures that data is delivered correctly.

5. Conclusion:
The selected packet belongs to the Transport Layer of the TCP/IP model because it contains TCP header information. It represents communication between my device and the Flipkart server while loading the website.